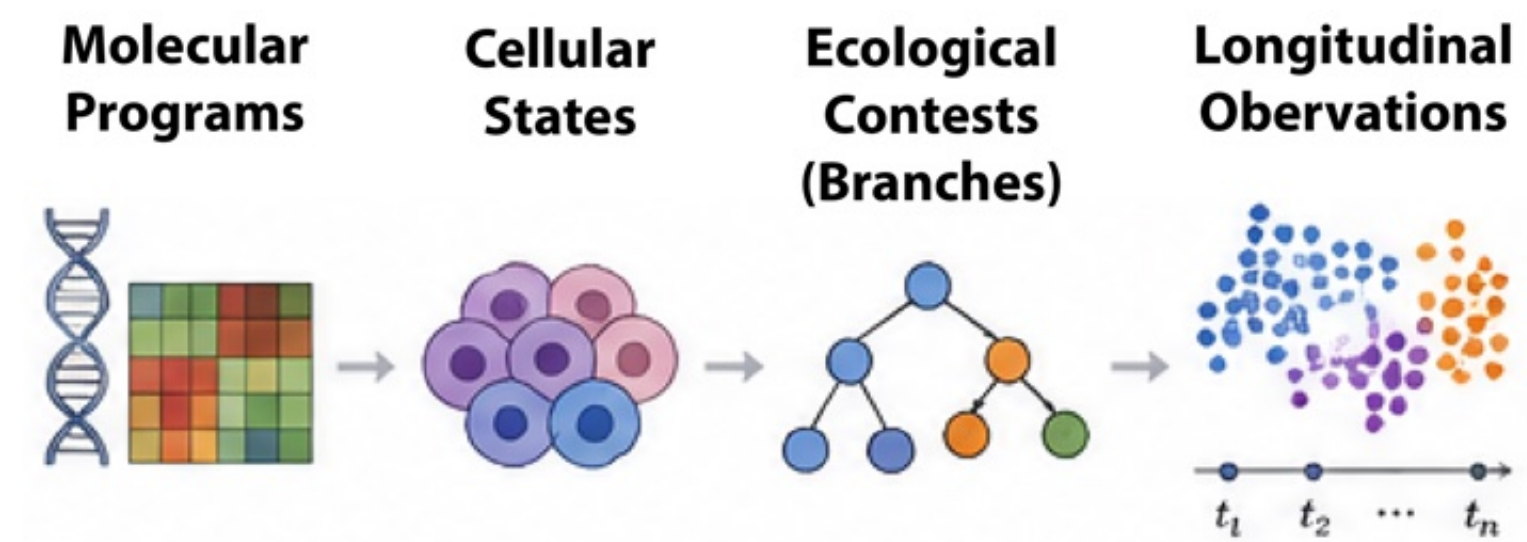


A generative probabilistic framework linking multiscale biological states, stochastic dynamics, branching evolution, and noisy observations

A Multiscale State Representation



Program activity (e.g., pathways, signatures) Cell states and compositions Lineages and ecological niches Spatial sites, time points, or donors with noise

Dimension p Dimension q Dimension k N observations

Latent $X_t \in \mathbb{R}^q$ evolves in continuous time; observations $Y_t \in \mathbb{R}^q$ are noisy projections.

B Unified Latent Process

The latent $X_t \in \mathbb{R}^q$ evolves as

$$dX_t = \Theta_{B_t} \{ \mu_{B_t}(t) - X_t \} dt + \Sigma_{B_t} dW_t + dJ_t$$

$\Theta_{B_t} \in \mathbb{R}^{q \times q}$: branch-specific restoring matrix (dynamic strength)

$\mu_{B_t}(t) \in \mathbb{R}^q$: branch-specific (possibly time-varying) attractor

$\Sigma_{B_t} \in \mathbb{R}^{q \times m}$: branch-specific diffusion matrix

$dW_t \sim \mathcal{N}(0, I_m dt)$: standard Wiener process

dJ_t : compound-Poisson Lévy jump component

Interpretation

Deterministic reversion toward an attractor + continuous diffusion + discontinuous jumps, with all components depending on the current branch B_t .

C Therapy-Dependent Attractor Shift

Attractors shift in response to therapy (and may vary over time):

$$\mu_{B_t}(t) = \mu_{0,B_t} + \Delta_{B_t} u(t)$$

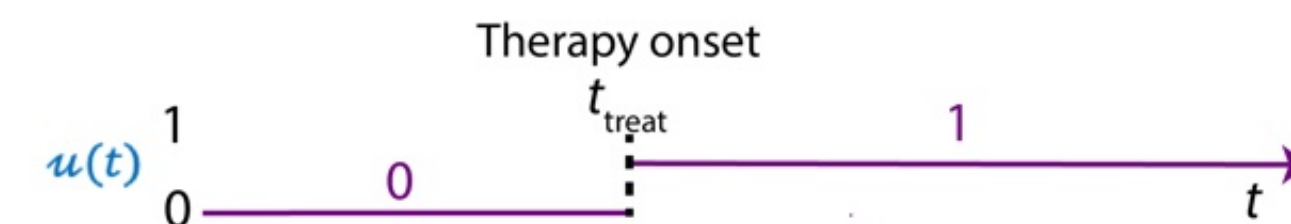
μ_{0,B_t} : pre-therapy attractor for branch B_t

Δ_{B_t} : treatment-induced shift for branch B_t

$u(t)$: treatment exposure function

$$u(t) = \begin{cases} 0, & t < t_{\text{treat}} \\ 1, & t \geq t_{\text{treat}} \end{cases} \quad (\text{step function})$$

or a time-varying dose/intensity

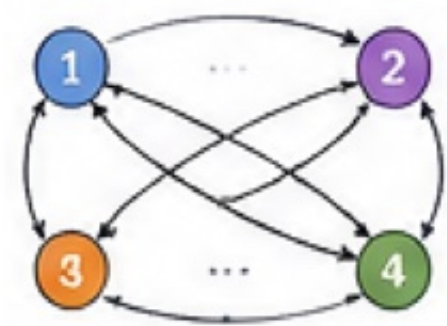


D Branch-State (Lineage) Process

Lineage identity $B_t \in \{1, \dots, K\}$ evolves as a continuous-time Markov chain with generator Q .

$$Q = \begin{bmatrix} q_{11} & \dots & q_{1K} \\ \vdots & \ddots & \vdots \\ q_{K1} & \dots & q_{KK} \end{bmatrix} \quad \begin{aligned} q_{kl} &\geq 0 \ (k \neq l): \\ &\text{transition rate } k \rightarrow l. \\ q_{KK} &= -\sum_{l \neq K} q_{Kl} \end{aligned}$$

Example ($K=4$)



Branch switches capture lineage transitions or ecological reorganization (e.g., niche shifts, state conversions).

E Observation Model

At observation time X_t we measure

$$Y_i = H_i X_{ti} + \varepsilon_i$$

$Y_i \in \mathbb{R}^{d_i}$: observation data at time X_t (e.g., scRNA-seq, spatial site)

$H_i \in \mathbb{R}^{d_i \times q}$: known or estimated projection/loading matrix

$\varepsilon_i \sim \mathcal{N}(0, R_i)$: observation noise

Allowing H_i, R_i to vary across sites, times, or platforms.

Consequences

We observe a noisy projection of the latent multiscale state. Estimation must account for irregular sampling and measurement error.

F Model Hierarchy and Special Cases

Our framework contains widely used models as special cases.

	Model	Characteristics	Obtained When
0	Brownian Motion (Drift)	$\Theta = 0, J_t = 0, \mu(t) = \mu_0$	No reversion, no jumps
1	Brownian Motion (Zero Drift)	$\Theta = 0, J_t = 0, \mu(t) = 0$	No reversion, no jumps, zero mean
2	Ornstein-Uhlenbeck (OU)	$\Theta > 0, J_t = 0, \mu(t) = \mu_0$	Reversion only
3	Shifted OU (Treatment-Response OU)	$\Theta > 0, J_t = 0, \mu(t) = \mu_0 + \Delta u(t)$	Reversion + treatment shift
4	OU with Jumps (OU-Lévy)	$\Theta > 0, J_t \neq 0, \mu(t) = \mu_0$	Reversion + discontinuous jumps
5	OU with Branching (OU-Branching)	$\Theta_{B_t}, \mu_{B_t}, \Sigma_{B_t}, Q, J_t = 0$	Reversion + branch switching
6	Full OULB (This Work)	$\Theta_{B_t}, \mu_{B_t}, \Sigma_{B_t}, Q, J_t \neq 0, \mu_{B_t}(t) = \mu_{0,B_t} + \Delta_{B_t} u(t)$	All mechanisms unified

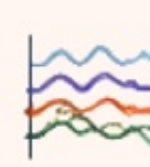
OULB provides a single coherent model that unifies continuous dynamics, treatment effects, jumps, and lineage branching under realistic observation.

Key Takeaway

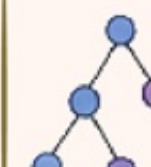
The OULB framework unifies multiscale states, therapy-driven attractor shifts, continuous stochastic evolution, jumps, lineage branching within a single interpretable generative model. By varying components, we recover classical models as special cases and gain a principled foundation for inference and prediction in complex cancer evolutionary systems.



Multiscale Latent State
 $X_t \in \mathbb{R}^q$



Reversion + Diffusion + Jumps



Branch Switching (Markov Chain)



Noisy Observations



Unified Inference and Prediction